

Skills**Languages:** C++, C#, Python, C, GML, HTML/CSS**Technologies:** Box2D, Arduino, Raspberry Pi, Git, Embedded Servers**Game Dev:** Custom engines, Unity, GameMaker**Core Concepts:** Developer Leadership, Custom Tooling, Image Processing Pipelines, Performance Optimization, Embedded Systems**Education****North Central Texas College**

- A.S. General Studies - 2024

University of North Texas

- B.S. Computer Science - Expected 2027

Projects**Campaign Companion** | *Founder & Lead Developer* | C#, Unity

- Founded **Hearthlight Studios LLC** to develop and commercialize Campaign Companion while directing a capstone development team.
- Led 2 concurrent student development teams across Software Engineering and Image Processing courses, with a third team onboarding for capstone (10 developers total). Earned highest course evaluation and selected as model documentation for future classes.
- Built custom C# tooling in Unity, including an interactive map-to-note node graph system, a custom image processing biome brush pipeline, and a rich-text Obsidian markdown parser supporting bitmap clipboard ingestion.
- Growing an active beta testing community of 78 current members on Discord through regular devlogs and alpha releases.

2D Fantasy Platformer | *Team Project* | C++

- Built a complete level pipeline from authored data to runtime rendering and Box2D physics within a custom C++ engine.
- Integrated LDtk level parsing and optimized rendering performance via camera-based tile culling.

Wireless DM-Controlled Magic Compass | *Embedded Systems Project* | C++, Arduino

- Developed C++ firmware and an embedded web server on constrained hardware for a wireless, servo-driven prop device.

Wireless DM-Controlled Datapad | *Embedded Systems Project* | Python, Raspberry Pi

- Designed handheld GUI software and a local web server with persistent storage on a Raspberry Pi for live gameplay management.

Portfolio

Scan to view Project Demos

